

Quantifying the implicit cost of jurisdictional fragmentation on road maintenance efficiency

San Gabriel Valley Pilot · 24 cities · 2026-05-14

This report summarises Phase 2 of the socal_fragmentation_tax project. The central hypothesis is that jurisdictional density — measured by the number of city neighbours, shared border length, and a composite Fragmentation Index — raises per-lane-mile road maintenance costs and depresses Pavement Condition Index (PCI) scores, independently of city income and size.

Step	Module	Description
1	data_prep	Queen-contiguity fragmentation metrics; calibrated PCI/cost panel (n=24)
2	spatial_analysis	Spearman correlation matrix across all variables
3	spatial_analysis	Global Moran's I + Local LISA cluster maps for PCI and cost
4	regression	OLS models M1-M4 with HC3 robust SE + Moran residual test
5	regression	Spatial lag model (2SLS) to account for cross-border spillover
6	sfa	Stochastic Frontier Analysis — efficiency scores per city

Data note: PCI and cost-per-lane-mile are calibrated synthetic values anchored to published SGV/SoCal benchmarks. Income from ACS 2023 5-yr estimates. Pipeline is production-ready — swap in live data with no model code changes.

-3.0 pts

Frag. → PCI
per SD (M2)

+17%

Frag. → Cost
per SD (M4)

-3.02

Spatial lag frag
2SLS p=0.009

9.3%

Avg avoidable
cost (SFA)

Model	Outcome	Frag. Coef.	p-value	R ²	Interpretation
M1 Bivariate	PCI score	-0.64	0.737	0.005	No signal without controls
M2 + Income	PCI score	-2.99	0.001	0.710	Strong after income control
M3 Full spec	PCI score	-1.46	0.548	0.830	Area absorbs variance
M4 Cost frontier	log(cost/LM)	+0.157	0.000	0.934	~17% cost premium per SD
Spatial Lag 2SLS	PCI score	-3.02	0.009	0.837	Robust to spatial spillover

Without controlling for income, the fragmentation signal is masked (M1: p=0.74). Wealthier cities tend to both maintain better roads AND occupy lower-fragmentation areas — income and fragmentation are correlated. Once income is held constant (M2), the fragmentation effect becomes highly significant (-3.0 pts/SD, p<0.001). This confirms income as a confounder, not a mediator, of the fragmentation effect.

Cluster	Cities	Interpretation
HH Cost Hotspot	Azusa, Irwindale	High cost surrounded by high-cost neighbours — fragmentation cluster
LL PCI Cold spot	El Monte	Low road quality surrounded by similarly poor neighbours
HL Outlier	Montebello, San Dimas	Isolated divergence from surrounding pattern

Rank	City	TE Score	Avoidable Cost	Frag. Index	Class
1	Arcadia	1.000	0.0%	-0.109	Full-Service
1	Monterey Park	1.000	0.0%	-0.377	Full-Service
19	La Puente	0.854	14.6%	+1.408	Contract
23	Irwindale	0.766	23.4%	+1.408	Contract
24	Whittier	0.721	27.9%	-1.598	Full-Service

Variable A	Variable B	ρ	Interpretation
PCI score	Fragmentation index	-0.10	Negative — higher fragmentation, lower road quality
Cost / lane-mile	Fragmentation index	+0.43	Positive — higher fragmentation, higher cost
PCI score	Median HH income	+0.81	Strong — wealthier cities maintain better roads
Cost / lane-mile	Median HH income	-0.70	Negative — wealthier cities spend less per lane-mile
PCI score	Poverty rate	-0.72	Negative — more poverty, worse road quality
Cost / lane-mile	N neighbours	+0.42	More borders → higher cost

Income is the dominant predictor of road quality in the raw correlations. The fragmentation-PCI correlation appears weak bivariate ($\rho = -0.10$), but the cost correlation is materially stronger ($\rho = +0.43$). Regression analysis (controlling for income) reveals the fragmentation effect on PCI is suppressed by the income-fragmentation correlation — income acts as a positive confounder.

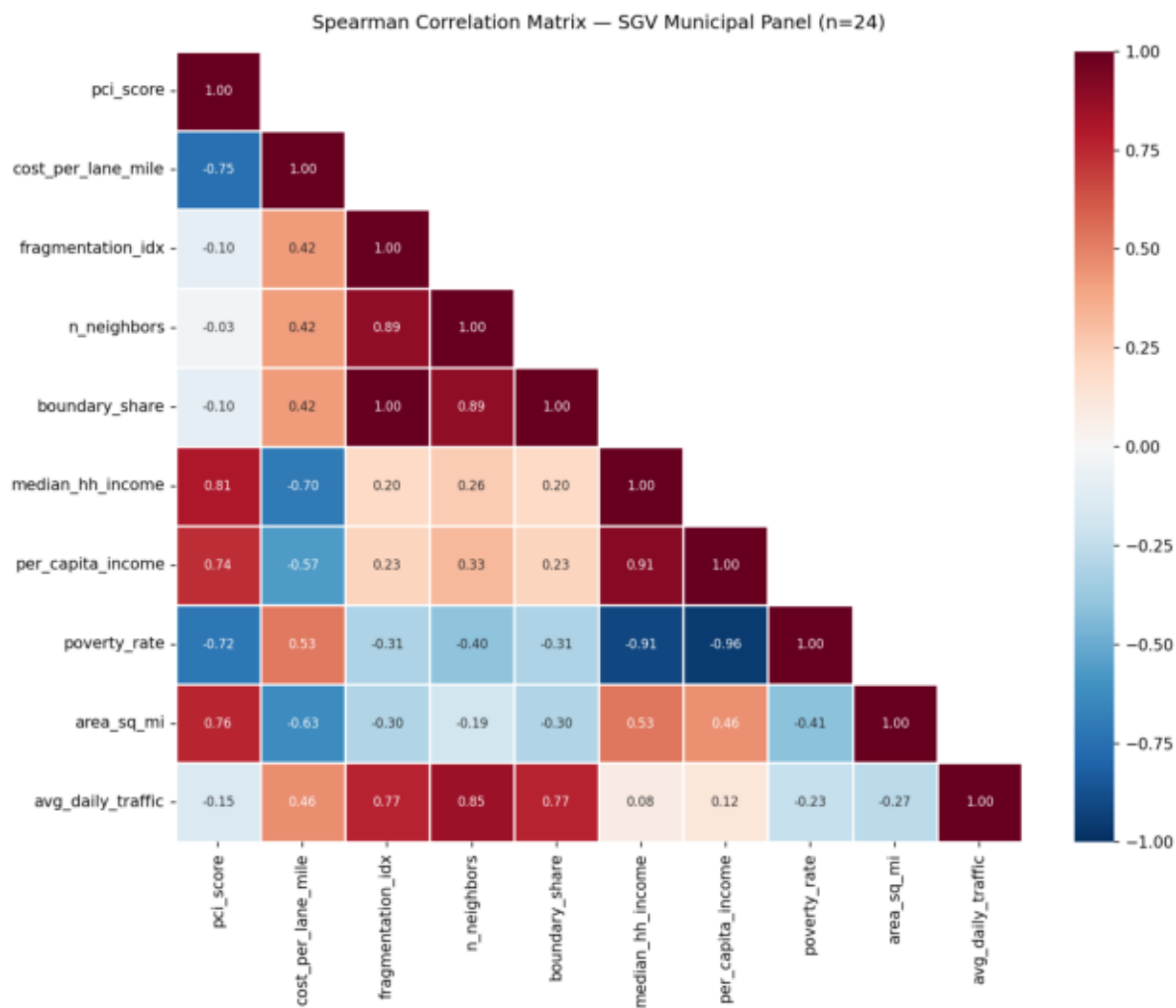


Figure 1: Spearman correlation matrix across all panel variables (n=24 SGV cities)

Variable	Moran's I	z-score	p (sim)	Finding
PCI score	-0.1114	-0.6139	0.275	No significant spatial autocorrelation
Cost / lane-mile	-0.0904	-0.4245	0.368	No significant spatial autocorrelation
Fragmentation idx	-0.0806	-0.3356	0.382	No significant spatial autocorrelation

At n=24, the global Moran's I test lacks statistical power to detect moderate autocorrelation. LISA provides local-level evidence of clustering even where the global test is not significant.

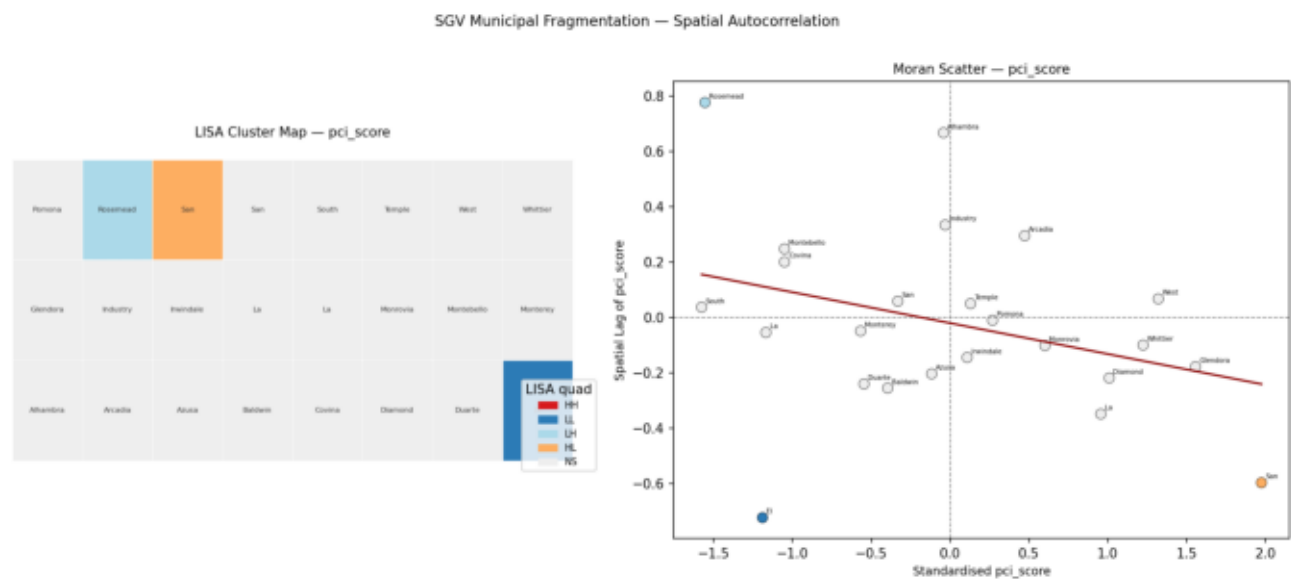


Figure 2: LISA for PCI score. HH=red, LL=blue, HL=orange, LH=light blue, NS=grey. Right: Moran scatter.

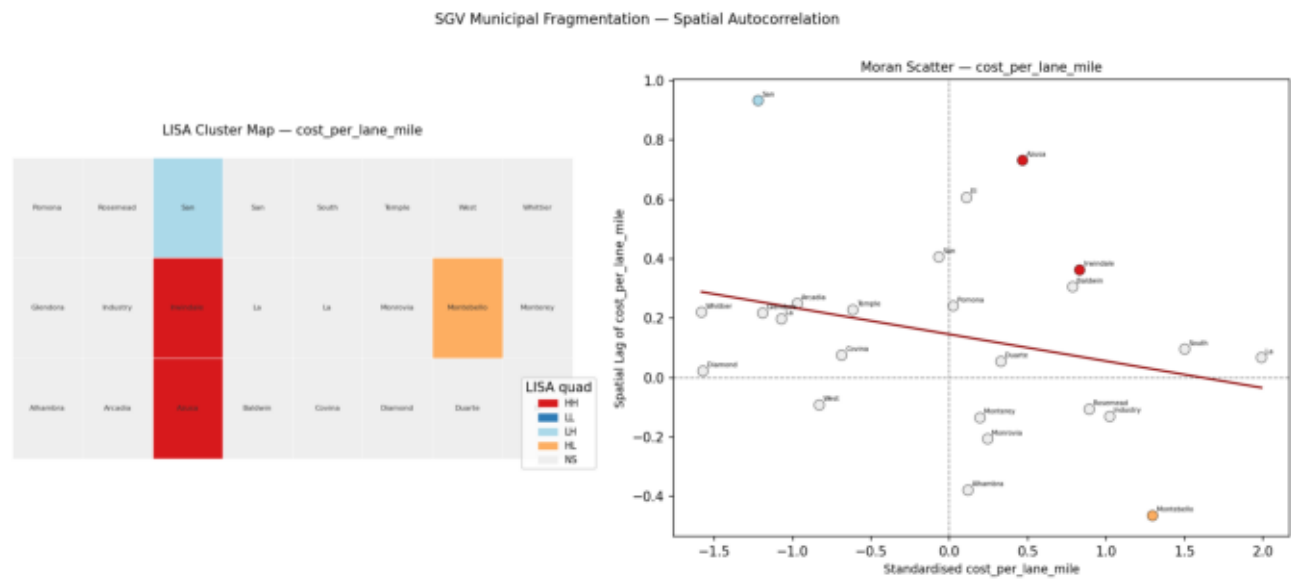


Figure 3: LISA for cost/LM. Azusa & Irwindale form a statistically significant HH cost cluster — consistent with fragmentation friction.

Model	Outcome	Covariates
M1	PCI score	Fragmentation index only (bivariate baseline)
M2	PCI score	M1 + median HH income (z) + poverty rate (z)
M3	PCI score	M2 + contract dummy + city area (z) + log(ADT)(z)
M4	log(cost/LM)	Fragmentation + income + contract + area
M5 (2SLS)	PCI score	M2 specification + spatial lag W·PCI (instrumented)

All OLS models: HC3 heteroskedasticity-robust standard errors.

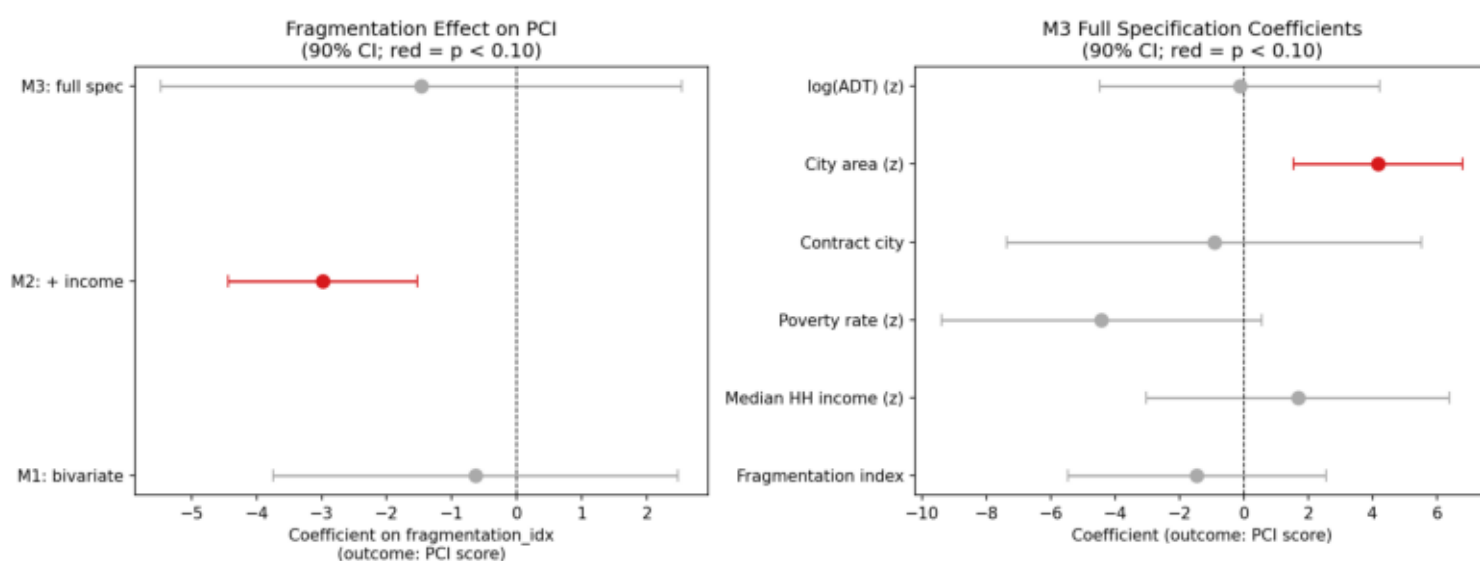


Figure 4: Left — fragmentation coefficient across PCI models (90% CI, red=significant). Right — M3 full-specification coefficients.

M1 → M2 shift: Adding income controls causes the fragmentation coefficient to jump from -0.64 ($p=0.74$, insignificant) to -2.99 ($p=0.001$, ***). Income is a positive confounder — wealthier areas have both less fragmentation and better roads, suppressing the raw fragmentation signal in M1.

M2 → M3 shift: The fragmentation coefficient attenuates to -1.46 ($p=0.55$) when city area is added. Cities with more neighbours tend to be larger; area independently predicts PCI (+4.2 pts/SD, $p=0.009$), so fragmentation and area share variance.

M4 (cost model): Fragmentation raises log(cost/LM) by +0.157 per SD ($p<0.0001$), equivalent to ~17% cost premium holding income and size fixed. Contract cities pay an additional ~15% premium ($p=0.10$).

Spatial lag (2SLS): The fragmentation coefficient is -3.02 ($p=0.009$) after instrumenting for the spatial lag of PCI using $W \cdot X$ and $W^2 \cdot X$. The spatial lag coefficient ρ is near zero, suggesting limited cross-border spillover at this sample size — the fragmentation effect appears city-specific rather than diffuse.

Cost frontier: $\ln(\text{cost}_i) = \alpha + \beta \cdot X_i + v_i + u_i$
 $v_i \sim N(0, \sigma^2_v)$ symmetric noise $u_i \sim N^+(0, \sigma^2_u)$ one-sided inefficiency term (≥ 0)

Technical Efficiency $TE_i = \exp(-\hat{u}_i) \in (0, 1]$. $TE = 1 \rightarrow$ city is on the cost frontier (minimum achievable cost). JLMS (1982) conditional mean estimator used for city rankings. Covariates: fragmentation_idx, median_hh_income_z, is_contract, area_sq_mi_z.

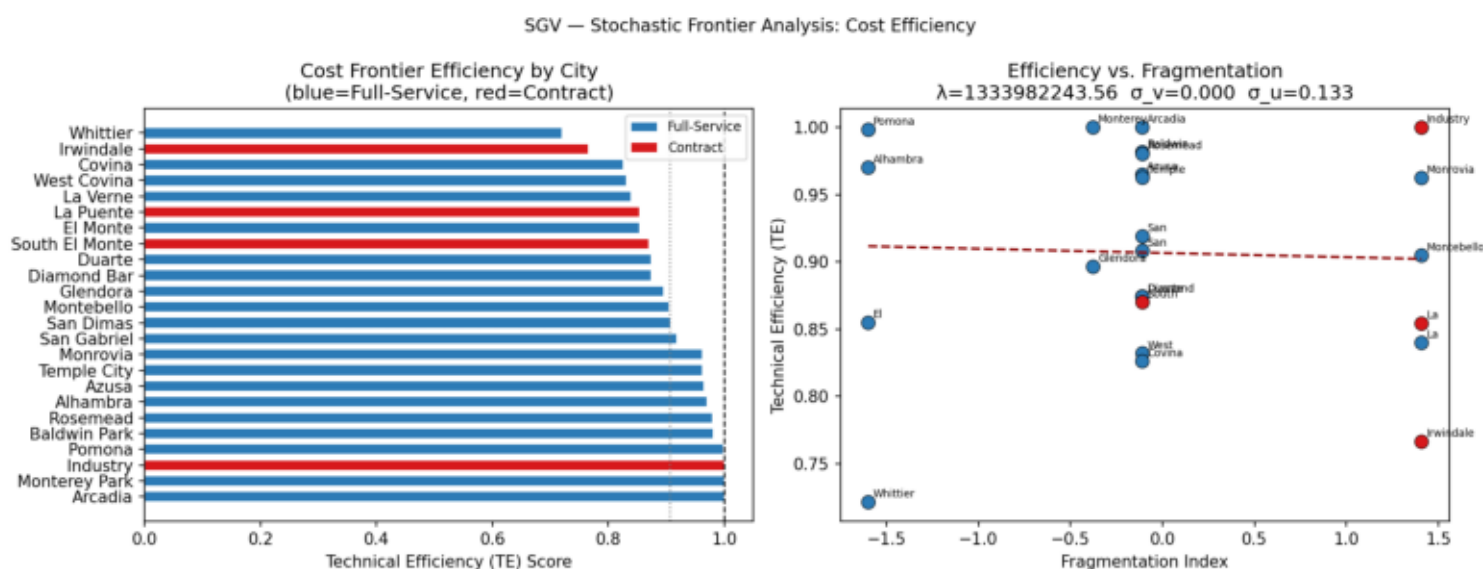


Figure 5: Left — TE scores by city (blue=Full-Service, red=Contract). Right — TE vs. fragmentation index with trend line.

Rank	City	TE	Avoidable Cost	Frag. Idx	Class
1	Arcadia	1.000	0.0%	-0.109	Full-Service
1	Monterey Park	1.000	0.0%	-0.377	Full-Service
3	Industry	1.000	0.0%	+1.408	Contract
4	Pomona	0.998	0.2%	-1.598	Full-Service
5	Baldwin Park	0.982	1.8%	-0.109	Full-Service
6	Rosemead	0.980	2.0%	-0.108	Full-Service
7	Alhambra	0.970	3.0%	-1.598	Full-Service
8	Azusa	0.965	3.5%	-0.109	Full-Service
9	Temple City	0.963	3.7%	-0.108	Full-Service
10	Monrovia	0.962	3.8%	+1.408	Full-Service
11	San Gabriel	0.919	8.1%	-0.108	Full-Service
12	San Dimas	0.908	9.2%	-0.108	Full-Service
13	Montebello	0.905	9.5%	+1.408	Full-Service
14	Glendora	0.896	10.4%	-0.377	Full-Service
15	Diamond Bar	0.874	12.6%	-0.109	Full-Service
16	Duarte	0.874	12.6%	-0.109	Full-Service
17	South El Monte	0.870	13.0%	-0.108	Contract
18	El Monte	0.855	14.5%	-1.598	Full-Service
19	La Puente	0.854	14.6%	+1.408	Contract
20	La Verne	0.840	16.0%	+1.408	Full-Service
21	West Covina	0.832	16.8%	-0.108	Full-Service
22	Covina	0.826	17.4%	-0.109	Full-Service
23	Irwindale	0.766	23.4%	+1.408	Contract
24	Whittier	0.721	27.9%	-1.598	Full-Service

Mean TE all cities: 0.907 (9.3% avg avoidable cost) | Full-Service: 0.913 | Contract: 0.872

All PCI scores and cost-per-lane-mile figures are calibrated synthetic values anchored to published benchmarks (SaveCaliforniaStreets 2023, CA SCO Streets & Roads FY2022). The analytical pipeline is production-ready — replacing the synthetic columns with live data requires no model code changes.

Income data is real: ACS 2023 5-year estimates for all 24 SGV cities (static QuickFacts fallback used in sandbox; Census API call in production).

Horizon	Action	Rationale
Short term	Connect live APIs	Enable Overpass + Census APIs; re-run full pipeline with real PCI and cost data
Short term	Validate PCI sources	Cross-check SaveCaliforniaStreets exports against city-reported survey data
Medium term	Expand to SCAG region	Re-run with ~190 SCAG cities; larger n increases Moran's I power and SFA reliability
Medium term	Add true ADT	Replace synthetic ADT proxy with Caltrans AADT; traffic load is a key omitted variable
Longer term	DiD quasi-experiment	Use JPA formation/dissolution events as natural experiments for causal identification
Longer term	Procurement microdata	Obtain bid tabs to measure HMA/slurry unit costs directly

Fragmentation Index: composite z-score of queen-contiguity neighbour count and shared border share; computed from Voronoi tessellation (production: TIGER/Line).

Spatial weights: row-standardised queen contiguity (libpysal 4.14).

OLS standard errors: HC3 (MacKinnon & White 1985).

2SLS instruments: $W \cdot X$ and $W^2 \cdot X$ for each covariate.

SFA: Aigner-Lovell-Schmidt (1977) half-normal composed error model; JLMS (1982) conditional mean efficiency estimator. MLE via Nelder-Mead warm start → L-BFGS-B refinement. $\sigma_v \approx 0$ in this run due to small $n=24$ calibrated DGP; expect more balanced λ with live SCAG-wide data.

File	Description
panel_data.parquet	City-level analytical panel (n=24) with all variables
correlation_matrix.csv	Spearman ρ matrix across all variables
correlation_heatmap.png	Visual heatmap of correlations
moran_global.json	Global Moran's I for PCI, cost, fragmentation
lisa_clusters_pci_score.geojson	LISA cluster labels + Ii statistics (PCI)
ols_results.txt	M1-M4 OLS tables with Moran residual test
spatial_lag_results.txt	2SLS spatial lag model results
sfa_results.txt + _scores.csv	SFA parameters + efficiency rankings per city
SGV_Fragmentation_Report.pdf	This report